

Span **XFRA**

A distributed AI data center in the backyard.

Span just announced a plan to put a 16-GPU AI inference pod — HVAC-sized, sitting on a concrete pad in the side yard — at **80,000 American homes**. **The hardware is real**. The economics are not yet proven.

This deck is the **strict reading** of what the evidence actually supports — across the technical claim, the unit economics, the strategic position, the risk surface, and the opportunities the analysis opens up. We build the argument one slide at a time. By the end, we get to **the three questions that decide the whole bet**.

NW OHIO

Northwest Ohio could be inside this footprint. Cheap PJM/AEP power, dense single-family housing across Toledo, Findlay, Lima and Bowling Green, expanding middle-mile fiber, and a workforce already absorbing data-center jobs from the New Albany corridor — the region has the inputs an 80,000-home distributed buildout actually needs. *The question for NW Ohio isn't whether the wave arrives; it's whether the region is on the host list or the bypass list.*

What if we used this technology in downtown Toledo?

Where this deck goes.

Twenty-one slides in two arcs — first the **strict analysis** of Span XFRA, then the **Northwest Ohio opportunity** that analysis opens up.

STRICT ANALYSIS · SLIDES 3–15

- 03 Executive summary — **the answer first.**
- 04 Technical viability — **start with the thing itself.**
- 05 Technical viability — **the orchestrator is the unknown.**
- 06 Business viability — **four numbers per node.**
- 07 Unit economics — **cost floor varies by market.**
- 08 Strategic positioning — **the moat is a combination.**
- 09 Competitive durability — **a queue is forming.**
- 10 Risk inventory — **not the ones in the press.**
- 11 Adjacencies 1 — **things Span could ship in 2027.**
- 12 Adjacencies 2 — **strategic shifts using the platform.**
- 13 Overall assessment — **strip it away.**
- 14 What we don't know — **the watchlist.**
- 15 In closing — **the bet is reasonable.**

NORTHWEST OHIO · CONCEPT · SLIDES 16–21

- 16 A **regional opportunity** worth taking seriously.
- 17 Build it — **the supply chain is here.**
- 18 Install it — **the trades are here.**
- 19 The use case **fits us better** than the Sun Belt.
- 20 What a **Toledo pilot** would look like.
- 21 The simple version — **somebody hosts it.**

↓ Click any line to jump · or use ← → to walk through.

Before the pieces, the answer.

Seven questions decide the whole thing. We walk through them in order — each one sets up the next. The pattern matters more than any single answer: confidence is high where the engineering is, and drops as you move toward economics and politics at scale.



Will the hardware work in the pilot?

Yes ●●● HIGH

Every component is **mature shipping product** — Dell PowerEdge chassis, Nvidia RTX PRO 6000 Blackwell, AMD EPYC, direct-to-chip liquid cooling. Integration is non-trivial but conventional. Of all the variables in the model, **the hardware itself is the smallest source of risk.**

What this means → Hardware that works on a bench still has to fit somewhere. The next question is whether a residential side yard can actually host it.

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Next →

Click any stop on the path · or use the buttons to walk the story

Start with the thing itself.

KEY A **quarter-million dollars** of conventional AI hardware in an outdoor HVAC-sized box. Nothing exotic — and that's the point.

The pod is a **conventional enterprise AI inference server** in an industrialized outdoor box. Roughly the size of an HVAC condenser. Every component ships today. **The expense is real — and it's that expense that forces the economic question on the next slides.**

GPUS 16 × Nvidia RTX PRO 6000 Blackwell Server Edition. ~\$9–10K each.	CPUS 4 × AMD EPYC. \$8.5–14K each.
MEMORY 3 TB DDR5. ~\$90–105K at DC-grade pricing.	COOLING Liquid Direct-to-chip. 35,000 BTU heat-pump rejection. ~60 dB.
BATTERY 15–16 kWh Whole-home backup; demand-response buffer.	SERVICE DRAW 19.2 kW 80 A at 240 V. IT load only ≈ 12.5 kW; rest is cooling + battery headroom.

↓ Total hardware: **\$310–400K per node**. At 80,000 nodes, that's **\$25–32B in compute capex** deployed across American suburbs.

The integration is solvable. The orchestrator is the unknown.

KEY Five components are **low or medium risk**. One — the orchestration layer (stop 5, in red) — is what could break the whole thing. Span has published **nothing** about it.

Walk the six technical components in order. Each stop is **color-coded by risk**: green for solved, amber for manageable, red for the unknown. The pattern — five steady stops and one red one — is the whole technical argument.



Hardware integration

Low risk ●●● HIGH

Dell PowerEdge chassis + Nvidia RTX PRO 6000 Blackwell + AMD EPYC. **Every part is shipping product**. The outdoor enclosure is industrialized HVAC engineering. Integration is non-trivial but well within Dell's and Span's combined competence.

Next → The box itself is solved. Next question: does it actually fit in a residential side yard?

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Next →

Click any stop · the red one (stop 5) is where the technical risk lives

The model reduces to four numbers per node.

KEY Three numbers are knowable today. The fourth — R, the revenue Span gets per node — is the one they haven't announced. Everything else cascades from it.

Span's whole pitch reduces to one piece of arithmetic. Three of the four numbers are observable today. The fourth — what a hyperscaler will pay — is the one Span needs to land and hasn't yet announced.

$$\text{Margin} = R - (C_{\text{pwr}} + C_{\text{net}} + C_{\text{op}})$$

R · REVENUE

Annual compute revenue from selling node capacity. The unknown.

C_{PWR} · POWER

Host's electricity bill, paid by Span directly to the utility.

C_{NET} · INTERNET

Host's internet bill, paid by Span to the ISP.

C_{OP} · OPS + CAPEX

Truck rolls, monitoring, insurance, amortized hardware.

↓ Plug in real numbers and the picture gets sharper.

Cost floor varies dramatically by market.

KEY NV/AZ at \$87K cost floor sits comfortably under the \$150K viability threshold. CA at \$111K sits much closer. The investor-implied ~\$200K target revenue sits above both.

In the Southwest pilot, the math has room to clear. In California — without a tariff carve-out — it doesn't. \$150K is the floor where this becomes a real business — costs need to land below it, revenue needs to clear it.

Southwest pilot
NV/AZ · \$87K total

Power \$17.5K

Internet \$1K

Ops \$7.5K

Capex amort (5yr) \$61K

California
\$0.30/kWh, no tariff · \$111K total

Power \$41K

Internet \$1K

Ops \$8K

Capex amort (5yr) \$61K

TARGET REVENUE

~\$200K
per node · per year

Back-solved from investor math. Abe Yokell (Congruent Ventures) framed each node as ~\$1M asset value — 50K nodes ≈ 1 GW × \$50B/GW Nvidia benchmark. Spread over five years with growth, that lands at \$200–250K/yr. Above the \$150K viability floor, well above the Southwest pilot cost — but Span hasn't announced what hyperscalers will actually pay.

Hover any chip above for the math behind each cost line.

The moat is a combination, not any single asset.

KEY No single asset is a moat. The combination of installed base + builder channel + power-control IP holds for 3–4 years. That's the window.

Assume the economics clear. The next question is defensibility. **None of Span's assets is individually a moat.** The combination — five years of installed panels, a top-three homebuilder channel, and the regulatory work to get power-control authority on a residential panel — is reasonably hard to copy in under three or four years.

Pulte partnership		Pulte can deploy with others; logo-strong but contractually unknown.
Smart panel installed base		5+ years of installs; real switching cost for builders standardized on Span.
Power-control IP & certification		Long process to UL-list and PUC-clear a panel with this authority.
Orchestration software		Pure software; replicable by any well-funded competitor.
Hardware design		Integration of commodity parts; no proprietary silicon.
Tariff & regulatory relationships		Slow to build, slow to copy.
First-mover narrative		Decays as competitors enter.

↓ Three to four years is the structural window. After that, defensibility depends on whether anchor contracts and tariff treatments lock in distribution.

And there's a queue forming.

KEY Telco edge is the most credible substitute. Span's bet: residential proximity and speed beat all three.

Three structural competitors, each attacking the same underlying problem — AI compute is power-limited, not silicon-limited — from a different angle. Telco edge is the most credible substitute. Crusoe Spark wins on density. Behind-the-meter-plus-flex has the capital.

TELCO EDGE

01

Verizon, AT&T, AWS Wavelength

Existing power, fiber, security at thousands of cell towers and central offices. Same distributed shape.

+ Operability & enterprise SLA

- Residential last-mile latency

MODULAR & ON-SITE POWER

02

Crusoe Spark + Edge Zones

Factory-built modular DC with diverse power options. 3-month time-to-power per unit.

+ Density, no residential friction

- Needs commercial parcel

BEHIND-THE-METER + FLEX

03

Bloom Energy + Emerald AI

On-site fuel cells + grid-flex orchestration. Bypasses interconnection queues entirely.

+ Capital available now (\$5B Brookfield)

- Higher capex per MW

↓ If competition is one half of the story, risk is the other.

The real risks aren't the ones in the press.

KEY Public commentary focuses on **noise and neighbors**, which are already managed at the spec. The risks that could actually **kill the business** — PUC tariffs, truck-roll scaling, anchor-tenant concentration, the orchestrator — get almost no airtime.

Press narrative lives in one place. Span's actual exposure lives somewhere else. **Here's the gap, side by side.**

WHAT THE PRESS COVERS

Already managed

- Neighborhood noise
Solved at the spec — 60 dB, below most municipal nuisance limits.
- NIMBY / zoning fights
Sidestepped by new-construction siting through Pulte.
- Aesthetic objection
Pod sized like an HVAC condenser. Municipal-compliant.
- "Backyard data center" framing
A marketing problem, not an engineering one.

WHAT ACTUALLY THREATENS THE BUSINESS

Unsolved, load-bearing

- PUC tariff denial **CRITICAL · MED-HIGH**
No state has ruled on residential headroom resale. The first ruling — likely NV or AZ — sets the precedent every other state references.
- Hyperscaler anchor walks **CRITICAL · MEDIUM**
Single-customer concentration risk at scale. The whole revenue model leans on landing one anchor buyer at premium pricing.
- Truck-roll cost scaling **HIGH · HIGH**
Pure scaling math — gets worse linearly with fleet size. Mitigated only by hardware reliability holding up over multi-year deployment.
- Orchestrator partial-failure cascade **HIGH · MEDIUM**
The unverified piece from slide 4. No public detail on how the system handles many simultaneous host outages.

↓ The left column manages itself. The right column is the watchlist for slide 13.

Things Span could ship in 2027.

KEY Four moves that **reuse the existing investment** — installed panel base, hardware envelope, orchestration software. No new business needed; just orchestration and packaging changes.

The first batch of adjacencies is the cheapest. The same platform Span is already running for XFRA can do **four other things** — selling demand response to utilities, packing more racks into a host site, capturing waste heat for hot water, and bundling its own risk into an insurance product.

NEAREST TERM

Utility grid-services revenue

Sell demand-response capacity to utilities alongside compute resale. The battery + panel + orchestration stack is already there. No incremental capex.

NEAREST TERM

Multi-rack site upgrade path

2–4 racks per non-residential host unlocks tightly-coupled workloads — training, larger inference. Same enclosure design; bigger pad.

MEDIUM-TERM

Heat reuse for residential hot water

12 kW IT load rejected as low-grade heat. In cold climates, 30–50% winter heating offset for hosts. Turns "free electricity" into "free electricity and free hot water."

MARGIN LEVER

Insurance-as-a-product

Glycol, cooling, and fire risk at residential sites creates a real bundled insurance product. 10–20% margin lift at fleet scale; also de-risks the host conversation.

↓ All four reuse what's already built. Next: shifts that reach entirely different customers.

Strategic shifts using the same platform.

KEY Four shifts that reach **different buyers, different geographies, different businesses** — all reusing Span's existing engineering, panel IP, and hardware envelope.

The second batch of adjacencies asks Span to reach beyond the residential-channel + hyperscaler-buyer model. Schools and civic buildings are a different buyer. The Northeast is a different market. The DOD is a different sales cycle. Telcos and retailers want the hardware without the service. **All four reuse the same engineering work.**

DIFFERENT BUYER

Sovereign / institutional channel

Schools, fire stations, hospitals, civic buildings. Host-as-customer model, not power-supplier. Sidesteps PUC risk entirely; lands government dollars.

DIFFERENT BUYER

Federal / DOD edge inference

On-base deployment skips PUC entirely. Sticky multi-year contracts, premium per-node revenue, different sales cycle than residential.

DIFFERENT GEOGRAPHY

Northeast expansion + heat reuse

High residential rates AND high heating costs. Pair XFRA with the heat-reuse adjacency and the rate-margin math becomes favorable. The Southwest pilot proves integration; the Northeast is where the economics get interesting.

DIFFERENT BUSINESS

Hardware-as-a-service

Sell the outdoor liquid-cooled enclosure as a product to telcos, retailers, and federal facilities — the Crusoe Spark play applied to Span's form factor.

↓ *Whatever the answer on XFRA itself, the platform underneath has surface area Span isn't yet using.*

Strip everything else away.

KEY Three questions decide it. **None is answerable today.** All three are answerable in the **next 18 months.**

The bet comes down to **three questions**. **None of them is answerable from public information today.** All three are answerable within the next 18 months. **Pilot data, the first PUC ruling, and the first hyperscaler contract** will resolve the whole thing.

01

Does the orchestrator deliver hyperscaler-grade SLA across a residential fleet?

02

Does at least one PUC, in at least one high-value market, approve a tariff that lets Span pay below retail for marginal kWh?

03

Does at least one hyperscaler sign a multi-year anchor contract at edge-premium pricing?

↓ A "yes" on all three makes XFRA a category-defining business with a 3–4 year window. A "no" on any one confines it to a profitable-but-bounded Southwest business — a real outcome, not the gigawatt ambition.

Watchlist, ranked by what would move the needle.

KEY Watch **tenant pricing** and **the first PUC ruling**. Everything else is texture.

If you're tracking this from the outside, this is what to watch. **The first two are decisive.** The rest become signal as the pilot reports out. The whole watchlist resolves **between now and end of 2027**.

1	First-tenant pricing and contract terms — who buys, how much, for how long, at what price.	DECISIVE
2	First PUC ruling on XFRA-relevant tariff. Likely Nevada or Arizona; sets precedent.	DECISIVE
3	Per-node operating cost from the pilot — truck-roll frequency, glycol events, failure rates.	HIGH
4	Orchestrator architecture and partial-failure behavior — public disclosure or independent audit.	HIGH
5	Host satisfaction and churn at month 12 of the pilot. Leading indicator for channel scale.	HIGH
6	Commercial XFRA spec and pricing (early 2027) — defines whether Span is residential-only.	MEDIUM
7	First major insurance carrier underwriting position on residential compute pods.	MEDIUM
8	Hyperscaler edge-strategy disclosures (AWS, Azure, Google). Could shrink addressable market.	MEDIUM

The bet is reasonable. The binary risks are public.

XFRA is a [credible, well-staged attempt](#) to convert a real structural inefficiency — residential grid underutilization — into a real economic asset: saleable AI inference capacity.

The [hardware works](#), the integration is solvable, the pilot is sized to test the right thing. [The thesis is not crazy](#).

But the binary risks — [orchestrator SLA, PUC tariff outcomes, anchor-tenant pricing](#) — are concentrated in places Span [has not yet pressure-tested in public](#). The next 18 months resolve most of the load-bearing uncertainty. [Track the three questions on slide 12; everything else is detail](#).

SPAN — XFRA press release (Apr 13, 2026)

Latitude Media — Can Span turn homes into AI inference hubs?

pv-magazine USA — Span and Nvidia AI data centers in your backyard

Network World — Startup SPAN teams with Nvidia

Data Center Richness — Liquid-Cooled GPUs Come to the Backyard

A regional opportunity worth taking seriously.

NW OHIO

Northwest Ohio has the **manufacturing base**, the **skilled workforce**, and (with heat reuse) the **climate profile** to turn an XFRA-type product into a regional industrial story — not just a deployment site.

Everything up to this point has been strict analysis. The next four slides shift mode — they ask **what an XFRA-style product could mean for Toledo and Northwest Ohio if we wanted to participate, not just observe**. Speculative, but concrete enough to act on.

↓ *Ahead: three regional advantages — we can **build it**, **install it**, and the use case **fits us better** — and then what a Toledo pilot might look like.*

First — we can build it.

NW OHIO The XFRA pod isn't exotic — it's **HVAC-form-factor metalwork** with liquid cooling and insulation. **Every layer of that supply chain lives within 30 minutes of Toledo.** Span ships pods from California; we could fabricate them here.

Most analyses of XFRA treat the deployment region as a passive host. **Northwest Ohio's industrial base would change that relationship.** The same supply chain that built the auto industry and the glass industry can build outdoor compute enclosures.

MANUFACTURING

Outdoor enclosure supply chain

The pod is steel-and-aluminum housing with insulation, liquid-cooling loops, and weatherproofing. Every layer has a regional supplier: **Owens Corning** (Toledo HQ — insulation, composites, glass-fiber reinforcement); **Dana Incorporated** (Maumee — thermal management, liquid-cooling expertise from decades of automotive driveline work); **regional metal fabrication** from the same shops that supply Stellantis Toledo Assembly; and **NSG/Pilkington** for any glass elements.

Span ships pods from California. We could fabricate them here at lower cost and shorter lead time.

SOLAR / PV ADJACENCY

First Solar at Perrysburg

North America's largest thin-film PV manufacturing facility sits 15 minutes south of Toledo. Recently expanded through a multi-billion-dollar investment, First Solar's Series 7 production keeps PV modules in the regional supply chain — not the global one. When XFRA pairs with behind-the-meter solar (the slide-10 optional add-on), the modules **don't need to cross an ocean** to reach a Toledo install.

Solar pairing becomes a regional supply story, not a logistics problem.

↓ *Building the pod is step one. Next: installing and servicing it.*

Second — we can install and service it.

NW OHIO

Building the pod is one thing. Getting it into 50 homes and **keeping it running for 5 years** is another. Toledo's electrical trades and homebuilder presence make both faster than fly-in installs from California.

A pilot is a logistics exercise as much as an engineering one. **The trades, the apprenticeships, and the homebuilder channel all exist here** — no fly-in crew required.

WORKFORCE

Skilled trades pipeline

Installing an XFRA pod needs three trades: **electrical** (200A service tie-in, panel work), **HVAC** (cooling loop, heat-pump rejection), and **plumbing** for the heat-reuse adjacency. **IBEW Local 8** has thousands of electricians in the Toledo area. Every neighborhood already has an HVAC tech who can work a heat pump. **Owens Community College** runs registered apprenticeships in all three trades. **University of Toledo** engineering and **BGSU College of Engineering and Innovation** produce the technical talent for orchestration and supervisory roles.

No fly-in California crew needed. The trades are already here.

BUILDER CHANNEL

Active homebuilder presence

Span's pilot model depends on builder pre-installation — the panel, conduit, and pad get installed **before move-in**. Toledo metro has active operations from **Pulte, NVR/Ryan, Drees**, and regional builders. They're building in Sylvania, Maumee, Perrysburg, Monclova, Oregon, and Holland. Volume is much smaller than Phoenix or Las Vegas — maybe 200–500 new builds per major builder per year — **but the relationship-and-process side of the channel exists and works.**

Lower volume means slower ramp, but also lower ops risk and faster iteration cycles.

↓ *Build it, install it. Next: why the use case actually fits us better than the desert.*

Third — the use case fits us better.

NW OHIO

The Sun Belt pilot has cheap power but **wastes the rejected heat**. We have a colder climate AND a latency hub within reach of **~30 million people**. Two structural advantages the Sun Belt can't claim.

The Sun Belt argument for XFRA is cheap power and new construction. **The Northwest Ohio argument is different — and arguably stronger**: heat that gets used instead of dumped, plus a population center that benefits from edge-inference latency.

CLIMATE

Cold winters turn rejected heat into a feature

A 12 kW IT load's rejected heat goes to waste in Phoenix — the cooling system just dumps it. In Toledo, that same heat becomes **a real co-benefit**. Captured into a pre-heat loop for the host's domestic water, it offsets natural-gas heating cost **estimated \$500–1,000/yr per node** depending on home size and gas rates. **The host signs up for the heat, not just the bill credit**. The slide-10 adjacency that's optional for Span becomes our primary value prop.

What's a thermal liability in Phoenix is a thermal asset in Toledo.

LATENCY

We're 30 ms from ~30 million people

Toledo sits at a Midwest crossroads — within ~30 ms RTT of Chicago, Detroit, Cleveland, Columbus, and Indianapolis. **That's roughly 30 million people** in the consumer-AI latency window for any inference workload that benefits from being close to end users — AR/VR, agentic chat, real-time recommendation, autonomous-vehicle inference. Span's Southwest pilot proximity argument works for the Sun Belt; **Toledo's argument works for the Great Lakes corridor**.

Edge inference for the Midwest doesn't need to live in Phoenix.

↓ Build it, install it, use it. Final slide: what a Toledo pilot actually looks like.

A Toledo pilot would look different — and that's the point.

NW OHIO

50-home pilot in Sylvania, Maumee, or West Toledo. Enclosures built locally. Heat reuse as the primary host value. Different inputs, different math, different story — one that includes a regional manufacturing angle Span's Southwest pilot can't tell.

Three differences from Span's Southwest pilot — and each one tilts in Toledo's favor.

DIFFERENCE 1

Local manufacture instead of CA ship

Outdoor enclosures fabricated from regional Owens Corning / Dana / metalwork supply chain. Cuts shipping cost, shortens lead times, creates a regional jobs story that local utilities and PUCs will reward in tariff conversations.

DIFFERENCE 2

Heat reuse as primary value

A 12 kW IT load rejects heat year-round. In Toledo winters, capturing it pre-warms the host's domestic water and offsets natural-gas heating cost — estimated \$500–1,000/yr per node depending on home size. Hosts sign up for the heat, not just the bill credit.

DIFFERENCE 3

Workforce-led install

Builder partner + IBEW Local 8 electricians + HVAC trades + AEP Ohio coordination. Local trades install faster than Span's CA crew can fly out — and the work stays in the local economy, which makes the political conversation easier.

WHO'S AT THE TABLE

Anchor partners to validate

First Solar (PV pairing). Pulte / NVR / Drees (builder channel). AEP Ohio or Toledo Edison (tariff conversation). University of Toledo PVIC (technical validation). BGSU College of Engineering and Innovation (engineering capacity + research partner). Owens Community College (workforce pipeline). Regional Growth Partnership and TMACOG (economic-development cover).

↓ This is concept, not commitment. Next step: a six-meeting roadshow — Pulte regional, AEP Ohio, First Solar BD, UToledo PVIC, BGSU College of Engineering and Innovation, and RGP — to see who'd put their name on a pilot.

If this works, somebody hosts it.

The only real question is whether that's **us**.

01

We can build it.

The enclosure supply chain — metal, insulation, thermal, glass — already lives within 30 minutes of Toledo. Owens Corning, Dana, First Solar.

02

We can install it.

IBEW Local 8 electricians, regional HVAC trades, and Pulte / NVR / Drees actively building in Sylvania, Maumee, and Perrysburg.

03

We can use it.

Cold winters turn the rejected heat into a **\$500–1,000/yr per node** value prop for the host — a feature the Sun Belt can't match.

NW OHIO

The pilot is six meetings away. Pulte regional · AEP Ohio · First Solar BD · UToledo PVIC · **BGSU College of Engineering and Innovation** · Regional Growth Partnership. *Either we set the table — or somebody else does, and we host their dinner.*